

ResolveIT - Smart Grievance & Feedback Management System (Hospital Use Case)

1. Introduction

In many hospitals, grievance handling is still managed manually, leading to delays, lack of transparency, and poor communication between patients and hospital staff. There is a need for a centralized digital platform that allows patients to submit grievances, track their status, and receive updates while enabling administrators to efficiently manage and resolve issues.

ResolveIT is a web-based Smart Grievance and Feedback Management System designed for hospitals. The system provides secure authentication, grievance submission, real-time status tracking, analytics visualization, and feedback management through an intuitive user interface.

2. Problem Statement

Hospitals often face difficulties in managing patient grievances due to manual processes and lack of centralized systems. Traditional methods result in delayed responses, poor transparency, and inefficient tracking of complaints. Patients are unable to monitor the status of their grievances in real time, leading to dissatisfaction. Administrators also lack analytical tools to identify recurring issues and evaluate resolution efficiency. Hence, there is a need for a secure and centralized digital platform to manage hospital grievances and feedback effectively.

3. Abstract

Hospitals frequently face challenges in managing patient grievances efficiently due to manual processes, lack of transparency, and delayed responses. The ResolveIT – Smart Grievance & Feedback Management System is a web-based application developed to digitally manage hospital grievances and feedback. The system enables patients to submit grievances and track their status, while administrators can monitor, update, analyze, and resolve issues through dashboards and analytics. The project uses modern web technologies to provide a secure, scalable, and user-friendly solution.

4. Objectives of the Project

The main objectives of the ResolveIT system are:

- To provide a centralized platform for hospital grievance management
- To enable patients to submit and track grievances easily
- To allow administrators to manage, update, and resolve grievances efficiently
- To visualize grievance status using analytics dashboards
- To improve transparency, accountability, and response time
- To reduce manual paperwork and improve operational efficiency

5. System Overview

ResolveIT follows a client–server architecture with a React frontend and a Spring Boot backend, connected through RESTful APIs.

User Roles

1. Patient

- Register and login securely
- Submit grievances with attachments
- View grievance status
- Submit feedback

2. Administrator

- View all grievances
 - Update grievance status (Pending → In Progress → Resolved)
 - View grievance analytics
 - Review patient feedback
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6. Technology Stack

6.1 Frontend Technologies

- React.js
- JavaScript
- HTML/CSS
- Chart.js

6.2 Backend Technologies

- Java
- Spring Boot
- RESTful APIs

6.3 Database

- MySQL

6.4 Tools

- VS Code
- Git & GitHub

7. System Architecture

The system uses a three-tier architecture:

1. Presentation Layer (Frontend)

- React components for UI
- State-based navigation
- Analytics charts and dashboards

2. Application Layer (Backend)

- Spring Boot REST controllers
- Business logic and validation
- File upload handling

3. Data Layer (Database)

- MySQL database
 - Tables for users, grievances, feedback, and files
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8. Frontend Module Description

8.1 Authentication Module

- Login and registration forms
- Role-based redirection (Admin / Patient)
- Local storage used to persist login session

8.2 Patient Dashboard

- View submitted grievances
- Track grievance status
- Submit new grievances
- Submit feedback
- Profile management

8.3 Admin Dashboard

- View total grievances
- View grievances by status (Pending, In Progress, Resolved)
- Update grievance status
- Access grievance analytics
- Review feedback

8.4 Grievance Analytics Page

- Visual representation of grievance status using Pie Chart
 - Summary statistics (Total, Pending, In Progress, Resolved)
 - Clean and professional UI for data interpretation
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9. Backend Module Description

9.1 Authentication Module

Handles:

- User registration
- User login
- Role assignment (ADMIN / PATIENT)

9.2 Grievance Management Module

Handles:

- Grievance submission
- File uploads
- Status updates
- Fetching grievances

9.3 Feedback Module

Handles:

- Feedback submission
 - Feedback retrieval by admin
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10. Database Design (Overview)

10.1 User Table

Field	Description
id	Unique user ID
full_name	User name
email	User email
password	Encrypted password

Field	Description
role	ADMIN / PATIENT

10.2 Grievance Table

Field	Description
id	Grievance ID
title	Grievance title
description	Issue description
department	Hospital department
status	PENDING / IN_PROGRESS / RESOLVED
created_at	Submission date
user_id	Patient reference
file_name	Attachment file

10.3 Feedback Table

Field	Description
id	Feedback ID
message	Feedback text
user_id	Patient reference

11. REST API Endpoints

11.1 Authentication APIs

Method	Endpoint	Description
POST	/api/auth/register	Register new user
POST	/api/auth/login	Login user

11.2 Grievance APIs

Method	Endpoint	Description
POST	/api/grievances/submit	Submit new grievance

Method	Endpoint	Description
GET	/api/grievances/all	Get all grievances (Admin)
GET	/api/grievances/user/{id}	Get grievances by patient
PUT	/api/grievances/{id}/status	Update grievance status
GET	/uploads/{fileName}	Download attachment

11.3 Feedback APIs

Method	Endpoint	Description
POST	/api/feedback/submit	Submit feedback
GET	/api/feedback/all	View feedback

12. Application Workflow

1. User registers or logs in
 2. System identifies user role
 3. Patient submits grievance
 4. Admin views grievance in dashboard
 5. Admin updates grievance status
 6. Status updates reflected in analytics
 7. Patient checks grievance status
 8. Patient submits feedback
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13. Features of the Project

- User Authentication and Role-Based Access:**
 Provides secure login and registration with separate access levels for patients and administrators.
- Online Grievance Submission:**
 Allows patients to submit grievances digitally with detailed descriptions and optional file attachments.
- Grievance Status Tracking:**
 Enables real-time tracking of grievance status such as Pending, In Progress, and Resolved.

- **Administrative Dashboard:**
Provides administrators with a centralized dashboard to view, manage, and update grievances efficiently.
 - **Grievance Analytics and Visualization:**
Displays grievance statistics using charts and summary cards to support data-driven decision-making.
 - **Feedback Management System:**
Allows patients to submit feedback and enables administrators to review it for service improvement.
 - **Dark and Light Theme Support:**
Enhances user experience by providing theme customization across the application.
 - **Secure RESTful API Communication:**
Ensures reliable and secure data exchange between frontend and backend systems.
 - **File Upload:**
Supports secure handling of grievance-related attachments through backend services.
 - **Responsive User Interface:**
The application is designed to be user-friendly and responsive across different devices and screen sizes.
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14. Security Features

- Role-based access control
 - Password encryption
 - Secure REST APIs
 - File upload restrictions
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15. Advantages of the System

- Improved grievance handling efficiency
- Transparent grievance status tracking
- Reduced paperwork
- Better communication between patients and hospital staff
- Visual analytics for decision-making
- Scalable and maintainable architecture

16.Conclusion

The ResolveIT – Smart Grievance and Feedback Management System successfully provides a digital solution for managing hospital grievances in an organized and transparent manner. By integrating modern web technologies with a clean user interface and robust backend services, the system enhances user experience, improves administrative efficiency, and supports data-driven decision-making through analytics.